

Data Analytics

Study Notes

PART 1: STATISTICS AND DISTRIBUTIONS

1. Uniform Distribution

A uniform distribution means every possible value has the same chance of happening. Think of rolling a fair die: you can get 1, 2, 3, 4, 5, or 6, and each has the same chance of 1 out of 6 (about 0.167 or 16.7%).

If you draw this on a graph, it looks like a flat rectangle because no value is more likely than another. The formula for the probability at any point is:

$$f(x) = 1 / (b - a)$$

Here, "a" is the smallest value and "b" is the largest value. Between these two points, the probability is spread out evenly.

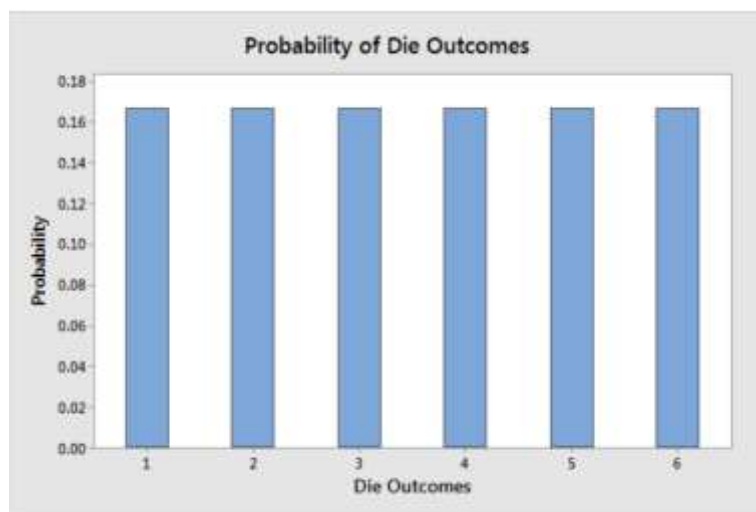


Figure 1: Bar chart showing equal probability (0.167) for each die outcome (1 to 6)

2. Left-Skewed Distribution (Negative Skew)

A left-skewed distribution is one where the long tail of the graph goes to the left side. Most of the data is on the right (higher values), but a few very low values pull the tail to the left.

This is also called negative skew. A simple example: if most students score very high on a test but a few score very low, the score data is left-skewed.

- The peak (most common values) is on the right side.
- The tail stretches out to the left.
- The mean (average) is usually lower than the middle value (median).

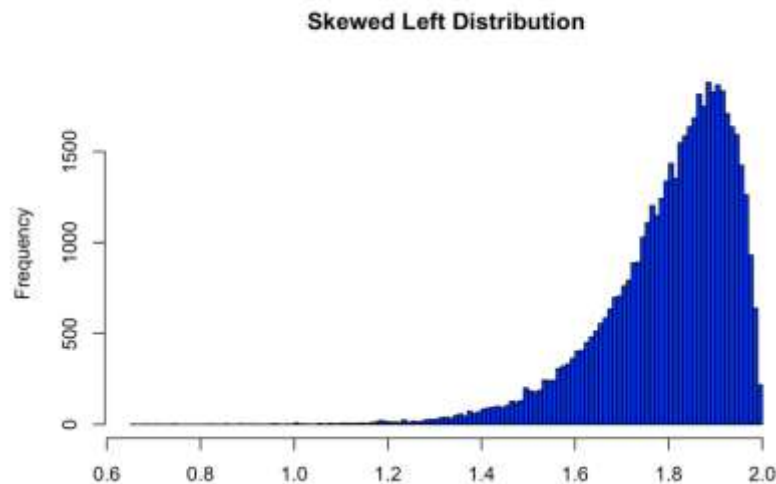


Figure 2: Left-skewed distribution - most data is on the right, with a long tail on the left

3. Normal Distribution (Bell Curve)

The normal distribution is the most common shape in statistics. It looks like a bell when drawn on a graph, which is why it is called a bell curve.

The key feature is that data is balanced perfectly on both sides of the middle (the mean). Values close to the mean appear most often, and values far from the mean appear less often.

- It is symmetric - the left side is a mirror of the right side.
- The mean, median, and mode are all at the same point (the center).
- About 68% of data falls within one step (standard deviation) from the center.

Example: The heights of 14-year-old girls, shown below, follow a normal distribution. Most girls are close to 1.512 meters tall, and very few are much shorter or much taller.

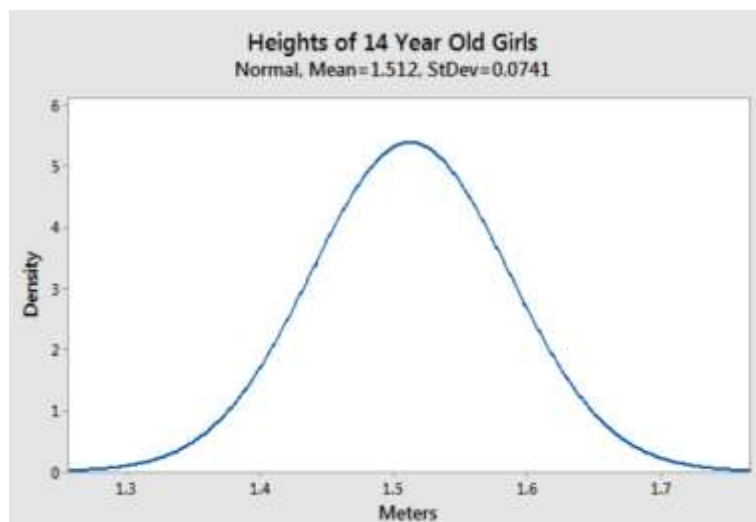


Figure 3: Normal distribution of heights of 14-year-old girls - a classic bell curve shape

4. Correlation

Correlation tells us whether two things (called variables) are related to each other. When one changes, does the other also change? There are three types of correlation:

4.1 Positive Correlation

When one variable goes up, the other also goes up. When one goes down, the other also goes down. They move in the same direction.

Example: As a person studies more hours, their exam score goes up. This is positive correlation. The correlation number is between 0 and +1. A value close to +1 means a very strong positive relationship.

High Degree of Positive Correlation

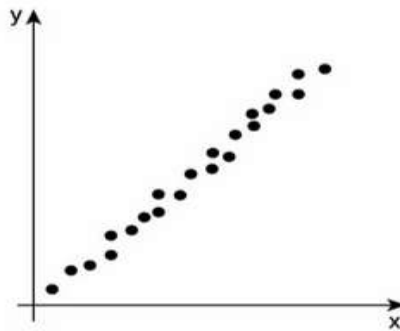


Figure 4: Positive correlation - dots move upward from left to right

4.2 Negative Correlation

When one variable goes up, the other goes down. They move in opposite directions.

Example: As the price of a product goes up, fewer people buy it. The correlation number is between 0 and -1. A value close to -1 means a very strong negative relationship.

High Degree of Negative Correlation

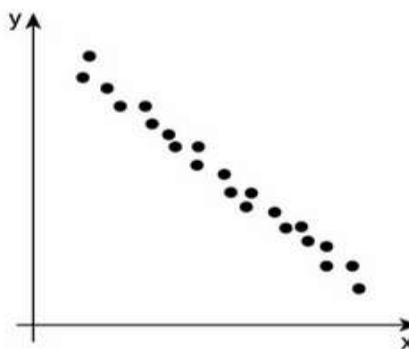


Figure 5: Negative correlation - dots move downward from left to right

4.3 No Correlation

There is no clear relationship between the two variables. One changing does not tell us anything about how the other will change. The dots on a graph appear scattered with no pattern.

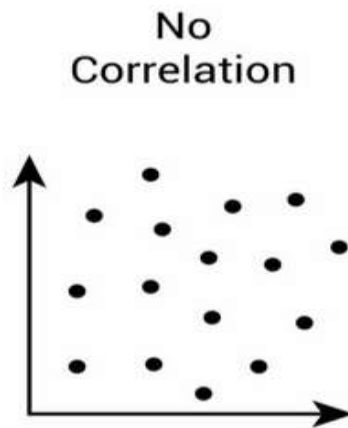


Figure 6: No correlation - dots are scattered randomly with no pattern

5. Linear vs. Non-linear Correlation

5.1 Linear Correlation

In linear correlation, the relationship between two variables follows a straight line. The linear correlation coefficient is a number that tells us how strong and in which direction this straight-line relationship is.

- A positive number means both variables increase together.
- A negative number means one goes up while the other goes down.
- A number close to 0 means there is little or no straight-line relationship.

5.2 Non-linear Correlation

In non-linear correlation, the two variables are related, but not in a straight line. The rate of change is not constant.

Example: When income increases by 20%, spending might increase by 5% sometimes, 10% other times, or 15% at other times. The change is not the same each time, which makes it non-linear.

PART 2: BUSINESS INTELLIGENCE AND ANALYTICS

6. Evolution of Decision Support and Analytics

The way businesses use data has changed a lot over the decades. Below is a simple timeline of how it developed from the 1970s to today:

Decade	Key Development	What Businesses Did
1970s	Routine Reporting, Decision Support Systems (DSS), Early AI & Expert Systems	Managers used reports and DSS to make decisions. AI started appearing.
1980s	Relational Databases, On-Demand Reporting, ERP Systems	Data storage became easier. Companies could request reports when needed. ERP integrated business processes.
1990s	Data Warehousing, Dashboards, Scorecards, Executive Information Systems (EIS)	Leaders could see summarized info at a glance. Warehousing allowed large-scale data storage.
2000s	Business Intelligence (BI), Cloud Computing, SaaS, Data Mining, Text Mining	BI tools grew. Cloud and SaaS made software accessible online. Mining techniques extracted insights.
2010s	Big Data, Advanced Analytics	Internet, social media, and smart devices created huge data. Advanced analytics helped process it.
2020s–Present	AI-driven Analytics, Machine Learning	Real-time predictive and prescriptive analytics guide decisions. AI automates insights.

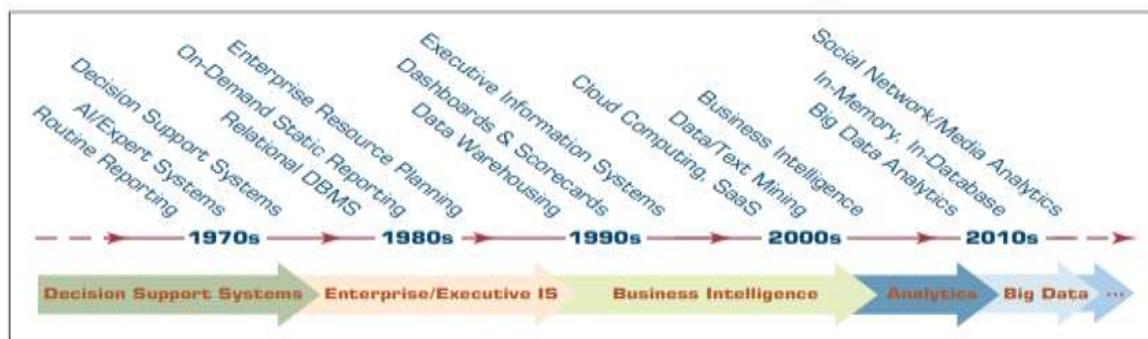


Figure 7: Timeline of evolution from Decision Support Systems (1970s) to Big Data Analytics (2010s)

7. The Rise of Big Data

Starting in the 2010s, the internet changed how data is created. New sources of data appeared everywhere: websites, social media, smart home devices, health monitors, and more.

This large wave of new data is called Big Data. It is called "big" not just because there is a lot of it, but also because it is fast, varied, and hard to handle with old tools.

To deal with Big Data, new technologies were created. For example, Hadoop with MapReduce allows processing of very large amounts of data across many computers at once. NoSQL databases store data in more flexible ways than traditional systems.

Social media is one of the most interesting and challenging new data sources. The data it produces is unstructured (not in neat tables or rows) and comes in very large amounts.

8. What is Business Intelligence (BI)?

Business Intelligence (BI) is a broad term that covers all the tools, systems, and methods used to collect, store, and analyze business data. The main goal of BI is to help people in a business make better decisions.

BI does this by:

- Giving decision makers easy access to data, sometimes in real time.
- Letting people explore and work with the data themselves.
- Turning raw data into useful information that leads to smart decisions.

The basic process of BI goes like this:

Data -> Information -> Decision -> Action

BI uses historical data (what happened in the past) and current data (what is happening now) to help managers understand situations and plan better for the future.

Assignment-01

- List three of the terms that have been predecessors of analytics.
- What was the primary difference between the systems called MIS, DSS, and Executive Support Systems?
- Did DSS evolve into BI or vice versa?

9. The Architecture of Business Intelligence

The BI architecture shows all the parts that work together to make Business Intelligence possible. Think of it as a map of how data travels from its source to a business decision.

Component	What It Means	Example
Data Sources	Where data comes from	Databases, spreadsheets, external systems
ETL (Extract, Transform, Load)	Process of collecting raw data, cleaning it, and loading into storage	Sales data cleaned and loaded into warehouse
Data Warehouse	Central storage for organized data	Company-wide data repository
Data Marts	Smaller sections of the warehouse for specific departments	Finance Data Mart, Sales Data Mart
Analytical Tools	Tools to query, analyze, and visualize data	OLAP, dashboards, reports, data mining
User Interface	What users see to interact with data	Browser, portal, dashboard

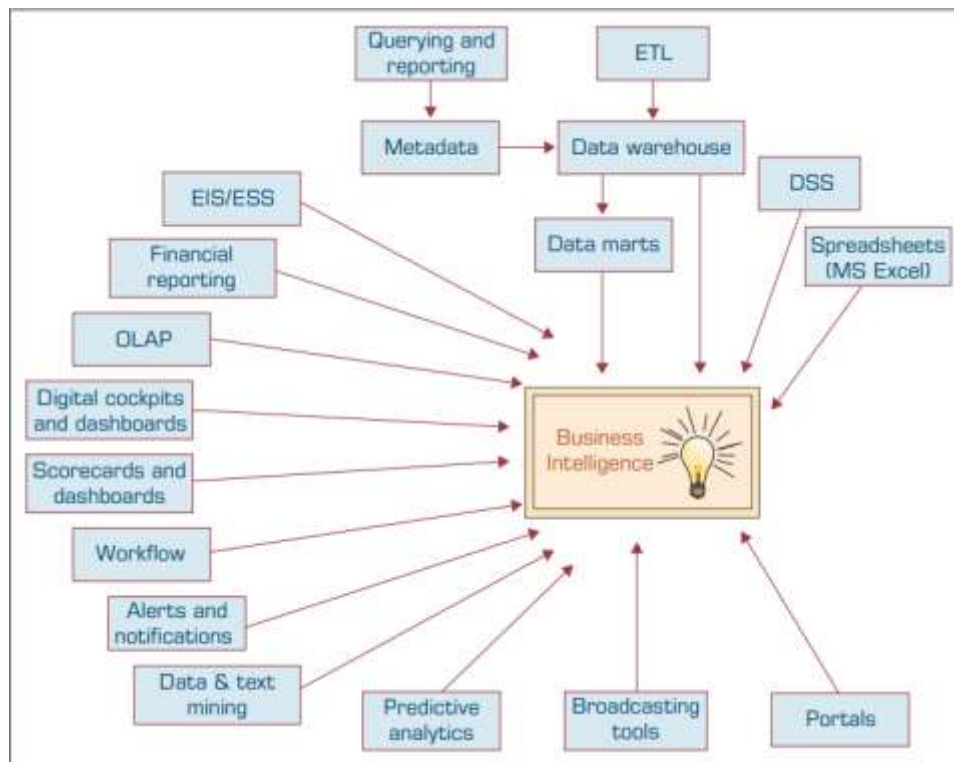


Figure 8: Architecture of Business Intelligence - how data flows from sources through the warehouse to business decisions

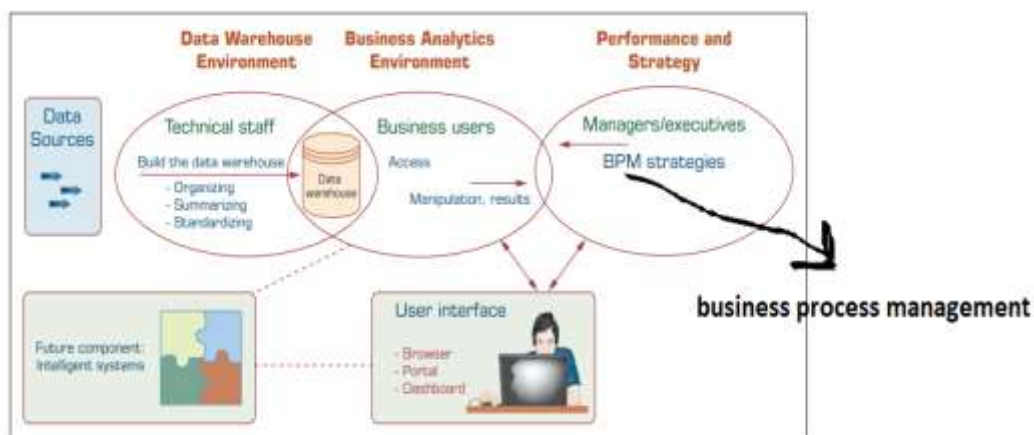


Figure 9: BI system showing Data Warehouse Environment, Business Analytics, and Performance and Strategy layers

Short Notes

Write short notes on:

- OLAP
- OLTP
- SCM/CRM

Case Study (Chapter 1): 1.1, 1.2, 1.3, 1.4, and 1.5

Self Study

- Is it a good idea to follow a hierarchy of descriptive and predictive analytics before applying prescriptive analytics?
- How can analytics aid in objective decision making?

10. Big Data: The 4 Vs

Big Data is described by four main characteristics, often called the "4 Vs":

V	Meaning	Explanation	Example
Volume	Amount	Very large amount of data, too much for old systems	Billions of Facebook posts
Velocity	Speed	Data arrives very fast, often in real time	Millions of tweets per minute
Variety	Diversity	Data comes in many forms	Text, images, videos, numbers
Veracity	Accuracy	Quality and reliability vary; some data may be noisy or uncertain	Fake news or incomplete records

Together, these four features create the challenge - and the opportunity - of Big Data. When handled well, Big Data connects to Business Intelligence and Analytics to create smarter decisions.

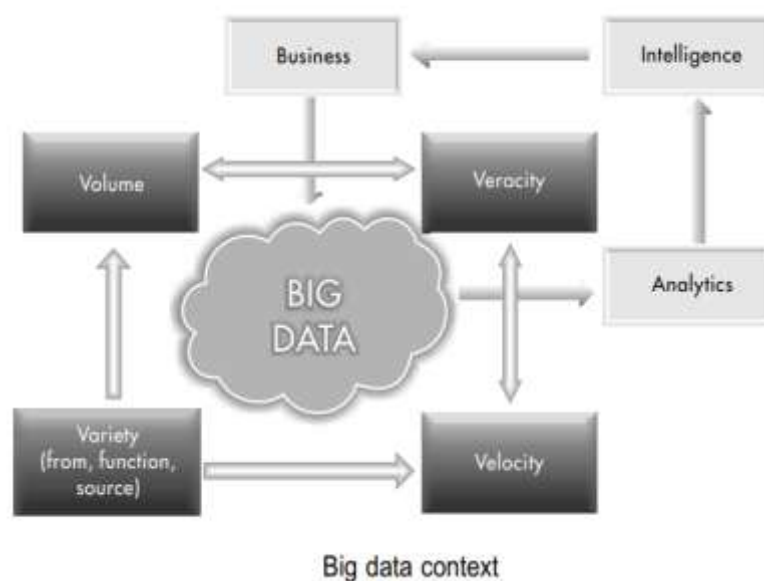


Figure 10: The Big Data context - Volume, Velocity, Variety, and Veracity feeding into Business Intelligence and Analytics

11. The Analytics Ecosystem

An analytics ecosystem is the full network of people, organizations, and tools that work together around a data-driven organization. It is not just about software - it includes many different players.

Component	What It Means	Role/Example
Analytics User Organization	The business using data	Center of the ecosystem
Data Generation Infrastructure Providers	Tools/systems that create and collect data	Sensors, websites, social media platforms
Data Management Infrastructure Providers	Keep data stored and organized	Database systems
Data Warehouse Providers	Store large amounts of cleaned data	Central repositories for analysis
Analytics-Focused Software Developers	Build tools for analyzing data	OLAP tools, dashboards
Application Developers	Create everyday business software	ERP, CRM systems
Academic Institutions & Certification Agencies	Train people and set standards	Universities, training centers
Regulators & Policy Makers	Create rules for data use	Government agencies
Analytics Industry Analysts & Influencers	Study and share analytics trends	Market research firms
Middleware Providers	Connect different systems with software	Integration tools
Data Service Providers	Offer data as a service	External data vendors

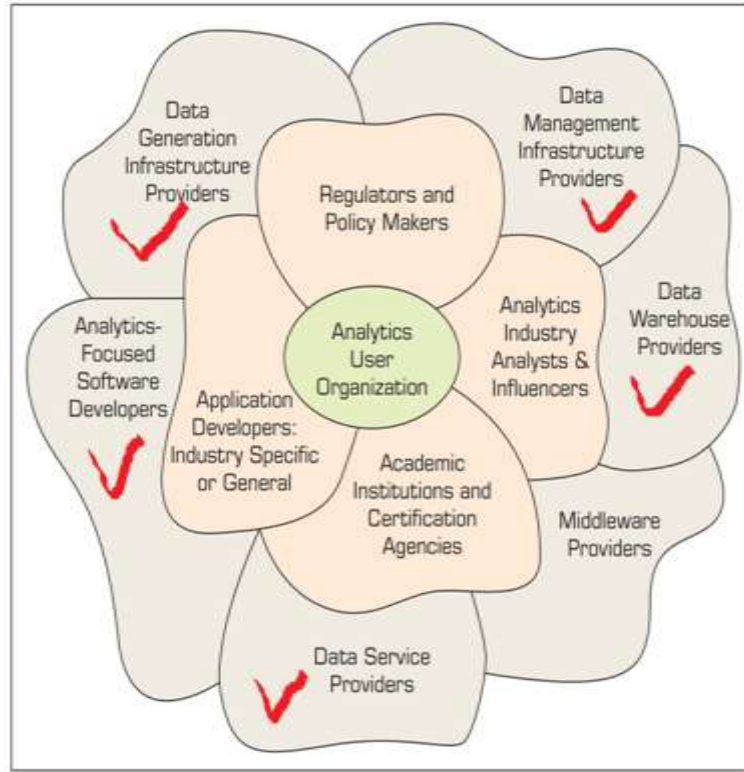


Figure 11: The Analytics Ecosystem - the network of organizations and providers surrounding an analytics user organization

অংশ	বাংলায় ব্যাখ্যা	কাজ
Analytics User Org	ডেটা ব্যবহারকারী প্রতিষ্ঠান	সিদ্ধান্ত নেয়
Data Generation	ডেটা তৈরি	সেন্সর, অ্যাপ
Data Management	ডেটা সংগঠন	ডেটাবেস
Data Warehouse	বড় ভাণ্ডার	বিশ্লেষণের জন্য
Analytics Software Dev	বিশ্লেষণী টুলস	BI Tools
Application Dev	ব্যবসার অ্যাপ	ERP, CRM
Academic & Certifiers	প্রশিক্ষণ ও মান	কোর্স, সার্টিফিকেট
Regulators	নিয়ম-কানুন	ডেটা আইন
Analysts & Influencers	ট্রেন্ড গবেষণা	রিপোর্ট, ব্লগ
Middleware	সংযোগকারী সফটওয়্যার	সিস্টেম লিঙ্ক
Data Service Providers	ডেটা সেবা	API, ডেটা ফিড

12. Overview: Business Intelligence and Analytics

The full subject of Business Intelligence and Analytics is organized into several major areas:

- Introduction (Chapter 1): Overview of Business Intelligence, Analytics, and Data Science.
- Descriptive Analytics (Chapters 2 and 3): Understanding what has already happened. Covers the nature of data, statistical modeling, visualization, and data warehousing.

- Predictive Analytics (Chapters 4 and 5): Predicting what might happen next. Covers data mining, text mining, web analytics, and social media analytics.
- Prescriptive Analytics (Chapter 6): Deciding what action to take. Covers optimization and simulation.
- Future Trends (Chapters 7 and 8): Big Data concepts and tools, future trends in analytics, privacy, and management considerations.

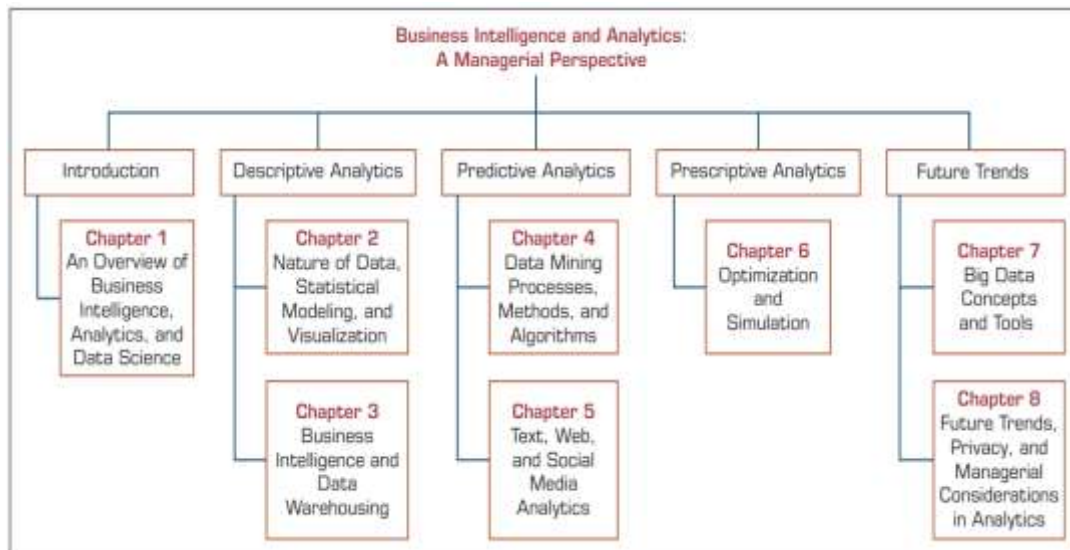


Figure 12: Structure of Business Intelligence and Analytics - from Introduction through Descriptive, Predictive, Prescriptive Analytics to Future Trends

Quick Summary

Part 1 covered how data can be spread in different patterns (uniform, skewed, or normal), and how two variables can be related to each other (positive, negative, or no correlation, in linear or non-linear ways). Part 2 covered how businesses use data to make decisions. Business Intelligence (BI) evolved from simple reporting in the 1970s to today's Big Data era, where massive amounts of fast, varied data are processed using advanced tools to help organizations act smarter and faster.